

Fine-Grained Bird Species Classification: A Comparative Study of Classical Descriptors and Deep Visual Embeddings on CUB-200-2011

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Abstract—Fine-grained visual categorization (FGVC) of bird species is a challenging computer vision task owing to high intra-class variance and subtle inter-class differences. This paper presents a systematic comparative study of feature extraction paradigms for classifying a 20-species subset of the Caltech-UCSD Birds-200-2011 (CUB-200-2011) dataset. We evaluate three hand-crafted descriptor families—Histogram of Oriented Gradients (HOG), Local Binary Patterns (LBP), and multi-colourspace Colour Histograms—both individually and in combination, pairing each with five classical machine learning classifiers (SVM, k-NN, Decision Tree, Random Forest, and Logistic Regression). We then replace hand-crafted features with 512-dimensional deep visual embeddings extracted from a frozen, pre-trained ResNet-18 backbone. Our feature ablation reveals that classical descriptors plateau at approximately 36% test accuracy, while the hybrid deep-feature pipeline achieves up to 79.55% accuracy with SVM—a gain of more than 40 percentage points—at a fraction of the computational cost of end-to-end fine-tuning. We further analyse per-class confusion patterns to identify the most and least discriminable species pairs.

Index Terms—Fine-grained classification, CUB-200-2011, bird species recognition, HOG, LBP, colour histograms, ResNet-18, transfer learning, support vector machine, feature ablation.

I. INTRODUCTION

FINE-GRAINED visual categorization (FGVC) involves distinguishing between visually similar sub-categories of a common class, such as different bird species, car models, or aircraft types. Unlike generic image classification, FGVC demands capturing highly localised and subtle discriminative visual cues (e.g., beak curvature, crown colour, wing-bar patterns) while remaining robust to variation in pose, lighting, and background clutter [1].

Bird species identification has practical applications in ecological monitoring, biodiversity conservation, and citizen-science platforms. State-of-the-art FGVC methods typically rely on end-to-end fine-tuned deep convolutional neural networks augmented with attention mechanisms [2, 3]. However, such approaches demand large labelled datasets, high-end GPUs, and substantial training time—resources often unavailable in resource-constrained environments.

In this work, we study whether a *hybrid* strategy—combining a frozen pre-trained CNN as a static feature extractor with lightweight classical classifiers—can close the accuracy gap relative to hand-crafted descriptors, without incurring the cost of full network fine-tuning.

Key Contributions:

- A systematic five-technique feature ablation study (HOG, LBP, Colour Histogram, Classical Combined, ResNet-18 Deep) evaluated under identical experimental conditions.
- A hybrid pipeline pairing frozen ResNet-18 deep embeddings with five tuned classical classifiers, achieving up to 79.55% accuracy on a 20-class CUB subset.
- Quantitative evidence that transfer-learned embeddings yield a >40 percentage-point accuracy gain over the best classical pipeline.
- A reproducible implementation with automated data-preparation pipelines and Colab notebooks.

II. RELATED WORK

A. Fine-Grained Visual Categorisation

Fine-grained recognition has been extensively studied on benchmark datasets including CUB-200-2011 [1], Stanford Cars [4], and FGVC-Aircraft [5]. Early approaches relied on part-based models and hand-annotated keypoints to focus on discriminative regions [6]. Bilinear pooling methods such as the Bilinear CNN [2] capture pairwise feature interactions. Recent

transformer-based approaches (TransFG [7]) improve discrimination via token selection and self-attention, albeit at high computational cost.

B. Classical Feature Descriptors

The Histogram of Oriented Gradients (HOG) [8] captures local edge structure and gradient distributions. Local Binary Patterns (LBP) [9] encode micro-texture via radial pixel thresholding. Multi-colourspace colour histograms [10] model chromatic properties across RGB, HSV, and CIELAB spaces.

C. Transfer Learning for Classification

He et al. [11] demonstrated that deep residual networks pre-trained on ImageNet learn generic, transferable visual representations. Using a frozen pre-trained CNN as a feature extractor combined with SVM or linear classifiers yields competitive accuracy with minimal compute [12, 13].

III. DATASET AND PREPROCESSING

The CUB-200-2011 dataset [1] contains 11,788 images across 200 bird species. We target a 20-species subset (~1,170 images) matching common avian families (e.g., Crow, Blackbird, Grebe, Gull). We apply a stratified 70/15/15% train/val/test split.

A. Bounding Box Cropping

Initial diagnostic runs showed low accuracy (~11–17%) due to background clutter. Parsing coordinates from bounding_boxes.txt and cropping each image before resizing focused descriptors on avian features.

B. Feature Standardisation

We apply StandardScaler (zero mean, unit variance) to the extracted feature matrices, ensuring high-magnitude dimensions do not dominate distance-based metrics.

IV. FEATURE EXTRACTION

A. Hand-Crafted Descriptors

- **HOG [8]:** Multi-channel HOG over grayscale, R, G, B channels with a spatial pyramid (1×1 + 2×2 grid), generating an 8,100-dimensional descriptor.
- **LBP [9]:** Multi-scale LBP combining fine (R=1, P=8) and coarse (R=3, P=24) textures, yielding a 10-d histogram.
- **Colour Histograms [10]:** 32-bin histograms across RGB, HSV, and CIELAB spaces, resulting in a 96-d descriptor.

B. Deep Visual Embeddings

ResNet-18 [11] pre-trained on ImageNet is stripped of its final classification head (replaced with an Identity layer), yielding a compact 512-dimensional semantic embedding for each bird crop.

V. CLASSIFIERS AND TUNING

Five classical classifiers were tuned on the validation set:

- (1) **SVM**: RBF kernel, $C \in \{1, 10, 50, 100\}$, $\gamma \in \{\text{'scale'}, \text{'auto'}\}$.
- (2) **k-NN**: $k \in \{3, 5, 7, 11\}$, distance-weighted voting.
- (3) **Decision Tree**: entropy/gini criteria, max depth $\in \{5, 10, 20, \infty\}$.
- (4) **Random Forest**: 100–500 estimators with depth bounds.
- (5) **Logistic Regression**: L2 penalty, one-vs-rest formulation.

VI. EXPERIMENTAL RESULTS

A. Feature Ablation Study

Table I presents SVM test accuracy across feature representations. HOG provides the strongest standalone classical baseline (33.52%). Combining classical descriptors reaches 36.36%, while ResNet-18 deep embeddings reach 79.55%.

TABLE I. Feature Ablation Study (SVM Test Accuracy)

Feature Representation	Dim.	Test Acc.
LBP Only	10	10.23%
Colour Histogram Only	96	15.91%
HOG Only	8,100	33.52%
Classical Combined (HOG+Color+LBP)	8,206	36.36%
ResNet-18 Deep Embeddings	512	79.55%*

B. Full Classifier Performance Comparison

Table II compares all five classifiers on classical vs. deep features. SVM achieves top performance on both representations, with deep features improving accuracy by over 40 percentage points across all models.

TABLE II. Comprehensive Model Benchmark (Classical vs. Deep)

Features	Classifier	Acc.	F1-Score
Classical	SVM (RBF)	34.66%	33.95%
	Random Forest	28.98%	28.51%
	Logistic Reg.	28.41%	27.55%
	k-NN	19.32%	16.81%
	Decision Tree	17.61%	17.84%
Deep	SVM (RBF)	79.55%	79.14%
	Logistic Reg.	76.14%	75.88%
	Random Forest	72.73%	72.41%
	k-NN	68.18%	67.10%
	Decision Tree	51.14%	50.87%

C. Error Analysis

Inspection of confusion patterns indicates that remaining errors under the deep-feature SVM model are concentrated between closely related species in the same genus (e.g., Rusty Blackbird vs. Brewer Blackbird), confirming that misclassifications reflect biological similarity rather than random artifacts.

VII. DISCUSSION

A. Why Classical Descriptors Plateau

Hand-crafted features encode low-level texture and edge statistics without semantic context. Furthermore, the 8,206-dimensional classical feature space causes sparsity given the dataset size (~820 training samples), limiting

generalization.

B. Why Deep Embeddings Succeed

Pre-trained ResNet-18 representations provide a dense 512-d manifold where intra-class variance is compressed and inter-class separability is maximized, enabling linear and RBF SVMs to establish clean separation boundaries.

C. Limitations

- **Scale**: Evaluated on a 20-species subset; full 200-class evaluation is planned.
- **Augmentation**: No geometric or photometric augmentation was applied during feature extraction.
- **Frozen Backbone**: End-to-end fine-tuning may yield further performance gains.

VIII. CONCLUSION

We presented a rigorous comparative study of classical and transfer-learned feature representations for fine-grained bird species classification. Hand-crafted features plateaued at 36.36% test accuracy, while frozen ResNet-18 deep embeddings achieved 79.55% accuracy with SVM—a 43-point improvement achieved with minimal compute overhead. This establishes a highly effective, accessible baseline for fine-grained avian recognition.

IX. REFERENCES

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